The impact of social networks on retirement behavior

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Abstract

We study how the effects of increases in the retirement age are amplified through social networks in the Netherlands. Between 2012 and 2025, the Dutch government gradually increased the statutory pension age (SPA) from 65 to 67. We implement an instrumental variable strategy and exploit these gradual increases in the SPA to identify the causal effect of social networks on individual retirement behavior. Using novel Dutch network data covering the entire Dutch population we use three types of networks, workplace, and family. We find significant evidence of social network effects increasing the pension age. A 10 percentage point increase in the share of retired peers causes an individual to reduce working hours by around 0.5 hours per month and increases the probability of retirement by around 0.3 percentage points. We find that in the workplace peer effects are driven by same- gender peers. Back of the envelope calculations show peer effects amplify the direct effect of an increase in the pension age with a 3% delay in retirement and 6% additional working hours.

JEL classification: J14, J22, J26, H55, D85

Keywords: retirement, statutory retirement age, social networks, peer effects, labor supply, older workers, Netherlands

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1 Introduction

Ageing populations and decreasing fertility rates are increasingly putting more stress on public pension systems. Age dependency ratios are predicted to increase from 31% in 2023 to 52% in 2060 and, as a result, public pension systems are unsustainable in the long run (OECD, 2025). To combat this, governments are continuously increasing the statutory pension age (SPA). However, the effectiveness of this instrument depends on how elderly workers respond to a higher SPA. Many studies show the importance of financial incentives created by institutional reforms in steering retirement behavior (Boskin and Hurd, 1978; Gruber and Wise, 2002; Coile and Gruber, 2007; Hanel and Riphahn, 2012). However, these financial incentives are unable to fully explain retirement behavior, partly because many people seem to retire at certain peak retirement ages. Lumsdaine, Stock, and Wise (1995) try to explain these peaks but are unable to do so using conventional retirement models. Instead, they bring forward the idea that social norms may play a role. According to the theory of conformity proposed by Bernheim (1994), social norms are transmitted through social networks since people want to conform to the norm of the group. Thus, understanding the role of social norms and networks in retirement behavior is of specific importance, as social network effects could amplify the effects of a pension reform by affecting not only individuals themselves but also their social network through them. Therefore, we investigate how the retirement behavior of social networks affects the labor force participation of elderly workers in the Netherlands. We focus on the Netherlands due to the availability of a rich administrative dataset containing carefully constructed social networks and the presence of gradual increases in the statutory retirement age, which provide us with a source of exogenous variation.

We define a social network as a group of individuals likely to interact or share similar characteristics. Social networks are available in three domains, namely, workplace, and family. As such, a social network effect appears when individuals are affected by the average behavior of peers from their social network. For example, if you work inside a company

where elderly workers retire only after the statutory retirement age, you might be inclined to postpone your retirement as well.

Within the analysis of social networks, two main problems arise. First, people might self-select into a certain social network based on unobservable characteristics and preferences. Second, within social networks, the direction of influence can be ambiguous (Manski, 1993; Duflo and Saez, 2002). We follow approaches from Oral, Rabaté, and Seibold (2026); Sturm (2025) that exploit exogenous variation in the retirement behavior of people within a social network. We use an instrumental variable approach where we instrument the retirement behavior of peers using their eligibility to claim state pensions. Due to gradual increases in the statutory retirement age for individuals depending on their birth cohort, we are able to identify the causal effect of a social network's retirement behavior on the labor force participation of the individual. Labor force participation is defined both on the intensive margin, hours worked, and on the extensive margin, retirement and labor market exit. Our identifying assumption requires that the birth cohort composition of an individual's peers does not determine the labor force participation of the focal individual other than through peers' retirement behavior.

We find modest, yet precisely estimated peer effects. More exposure to retired peers induces individuals to reduce their working hours, increases their chance of retiring and reduces their probability of working. These results hold across both the workplace and family specific network samples. Our results are robust to numerous alternative specifications and a placebo test shows that our estimates cannot be reproduced using randomized peer groups, further supporting a causal interpretation of the peer effects we identify.

In the next section, we contextualize ourselves within the broader literature on social network effects. Following this, section 3 discusses the institutional setting of the pension system in the Netherlands. Next, section 4 discusses the data and shows descriptive figures and tables. Subsequently, we discuss our method for identifying the causal effect and how we plan to model this in section 5. In section 6, we discuss the results and show that our

results are robust to numerous robustness tests. We provide interpretation and discussion for our findings in section 7. Finally, we conclude the paper in section 8.

2 Literature

This section provides an overview of the relevant literature on social network effects and labor force participation. First, we discuss our contribution to the literature on peer networks and retirement behavior. Subsequently, we position ourselves in the literature on the labor market effects of increases in statutory retirement ages.

The literature on social network effects and retirement behavior starts with two papers by Duflo and Saez (2002, 2003). They study the importance of workplace peer effects in a university for decisions regarding retirement savings. The first paper uses an instrumental variable approach, followed by the use of a randomized experiment in the second paper. In both papers, the authors find a significant role for peer effects affecting retirement savings decisions. More related to labor force participation is the paper by Vermeer, Van Rooij, and Van Vuuren (2019), who use survey data to determine the role social networks play in the retirement preferences of workers. They find an increase in the preferred retirement age by 3 months when workers' social network increases the retirement age by one year. This effect is more pronounced when workers initially preferred to retire before the SRA, indicating a norm effect. However, since retirement preferences may differ from actual retirement behavior, a body of papers exploits exogenous pension reforms to study peer effects on retirement behavior using administrative data (Brown and Laschever, 2012; Sturm, 2025; Badalyan, 2025; Oral et al., 2026). The paper by Brown and Laschever (2012) focuses on teachers and examines the effect of peers on a teacher's likelihood of retirement. They find that an increase in the number of retired peers increases teachers' own retirement probability by 1.5–2 percentage points. Sturm (2025) and Badalyan (2025) both exploit a German pension reform which increased the early retirement age for women from 60 to 63. Both papers find

that women are more likely to postpone their retirement when, in response to the reform, their peers similarly extend their working lives. However, the papers above focus on peer effects in the workplace, while other social networks might reveal peer effects as well. The paper by Oral et al. (2026), most relevant for our paper, examines the network effects on retirement for three different network levels, namely: neighborhood, workplace, and family. They exploit a major pension reform in the Netherlands, the 2005 abolishment of the VUT, which created a large disincentive for older workers to exit the labor force through early retirement. They find substantial spillovers of the pension reform between similar individuals within the same network, but note that these effects become much smaller for the average peer. Our contribution to the above literature is threefold. First, we use a high-quality administrative dataset containing carefully selected networks on three levels. We believe our identification of social networks is more careful as we restrict the degree of homogeneity within a network. Second, we study the network effects of a continuous pension reform which gradually increases the SRA. Since increases in the SRA are common in many international contexts and are expected to remain the preferred instrument of governments, we argue that our results have important policy implications. Finally, we examine the network effects on labor force participation both on the intensive and extensive margin. Moreover, we study the occurrence of substitution with other social security programs.

Another mechanism through which pension reforms induce spillover effects on labor force participation is through partners. In other words, does a change in the retirement age of the older partner affect the labor supply of the younger partner? Across multiple developed OECD countries, papers find the presence of spillover effects of pension reforms between partners (Atalay, Barrett, and Siminski, 2019; García-Miralles and Leganza, 2024; Rutten, Knoef, and van Vuuren, 2023). One of the proposed mechanisms for these spillover effects is the presence of leisure complementarities. While this mechanism is likely to play a smaller role in the broader social networks in this paper, it might help to explain some of the spillover effects.

A large share of the literature focuses on the direct effects of pension reforms that increase the normal retirement age (NRA) or the early retirement age (ERA) on elderly labor supply. For example, focusing on extensive margin effects, Mastrobuoni (2009) finds that an increase in the NRA led affected workers to postpone retirement by about half as much as the increase in the NRA. Using increases in the ERA in Austria, Staubli and Zweimüller (2013) find an increase in employment of around 9.75 percentage points for affected men and 11 percentage points for affected women. In the Dutch context, Rabaté, Jongen, and Atav (2024) examine the effects of the gradual increases in the SPA between 2012 and 2019, and find large employment effects of the reform. The latter two papers also investigate whether retirement age increases induced substitution into other social insurance programs. These results are inconclusive: whereas Staubli and Zweimüller (2013) find large flows to unemployment insurance in Austria, Rabaté et al. (2024) do not find any substitution into social insurance after the increase in the SPA in the Netherlands. "‘

3 Institutional setting

We focus on social norms affecting retirement behavior within the context of the Netherlands. In this section we explain the institutional setting in which the Dutch make retirement decisions. We discuss the pension system and its reforms and provide an overview of other social insurances which might provide an alternative path into retirement.

3.1 Dutch pension system

The Dutch pension system consists of three pillars. The first pillar comprises the public pension benefits funded by workers through a pay as you go system. Individuals can claim these benefits upon reaching the SRA and receive an amount roughly equal to 70% of the minimum wage. For individuals living with their partner this amount is 50% of the minimum wage. Individuals receive this pension relative to the number of years they reside in the

Netherlands, where each year corresponds to 2% of the full benefits. The age at which individuals can claim these benefits depends on the year they reach the current SRA, and thus on their birth year. The SRA was 65 until 2013, after which it gradually increased to 67 in 2024. From 2022 onward, the Dutch government adopted a new law linking the SRA to life expectancy, meaning that for every 4.5 month increase in life expectancy the SRA increases by 3 months. The government now announces the specific SRA age 5 years in advance. Figure 1 below summarizes the specific SRA's over the period studied in this paper.

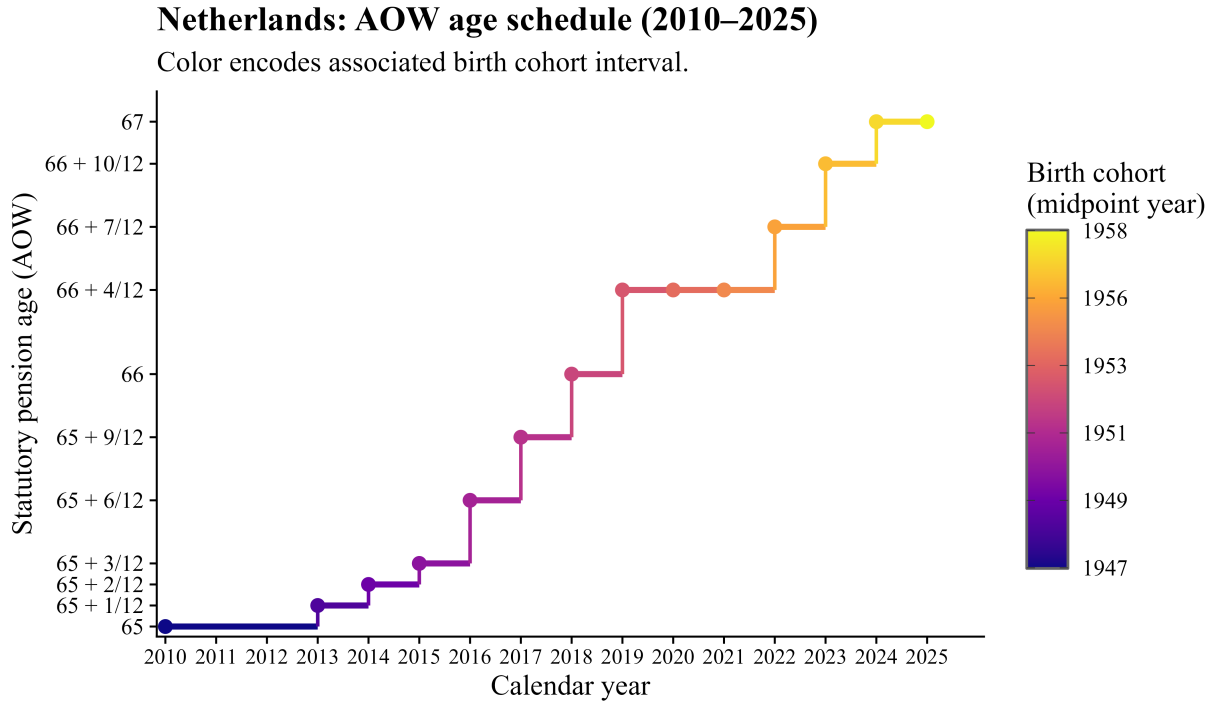


Figure 1: Netherlands: AOW age schedule (2010–2025).

The second pillar of the Dutch pension system consists of occupational pensions. Employees build up supplementary pension benefits through their employer, with contributions made by both the employee and the employer. These pensions are administered and held by pension funds or insurance companies. The size of the second pillar pension depends on an employees tenure and yearly earnings. In contrast to the first pillar pensions employees can claim the second pillar pension before the SRA.

Finally, the third pillar consists of supplementary pensions such as annuities, life insurances, or bank savings.

3.2 Pension reforms

Besides the gradual increases in the SRA, the Dutch pension system knows multiple reforms. The most drastic change happened between 2005 and 2006 when the government phased out the VUT and pre-pension, two generous early retirement schemes. From then on, workers could still retire before the SRA but the reform made it significantly less attractive to do so.[1] However, this reform also introduced a new arrangement, "Levensloop Regeling" in Dutch, which allowed workers to save with favorable conditions, and to use these savings as means to retire early. From 2012 this arrangement was not possible anymore leaving no attractive way for early retirement until a new reform in 2021. This reform removed the large levy employers have to pay when a worker retires before the SPA. Because these reforms may affect retirement behavior, we include cohort and collective labor agreement fixed effects to account for differences in retirement conditions across cohorts and agreement-specific retirement rules.

4 Data

4.1 Data source and sample selection

We use administrative data from Statistics Netherlands. Statistics Netherlands provides, under strict conditions, a broad set of data on the dutch population and companies. We collect and compile data from three main categories within the infrastructure of Statistics Netherlands: 1) individual level data on employment history, including social benefits, 2) individual level characteristics and demographics, and 3) network data for individuals at two different levels.

[1] In their paper, Oral et al. (2026) use this reform to study the network effects of pension reform.

We start by selecting a cohort of individuals born between 1950 and 1959. This will be our baseline population and contains roughly 2.75 million individuals. The 1950 cutoff reflects the large reform of the Dutch early-retirement system in 2005 discussed in section 3.2: individuals born on or after 1 January 1950 are excluded from the transitional arrangements preserving access to generous VUT benefits. We collect the social networks and employment histories of our sample in the period 2010-2024 and construct network specific analysis samples depending on whether the focal individual has a valid family or workplace network at the age of 59. This means we end up with two distinct samples, originating from the same baseline population, one sample for individuals with workplace peers and one with family peers.

4.2 Outcome variables and definitions

We define our extensive margin outcome variables based on the main source of income in a specific month. Sources of income include work, pension, social assistance, unemployment benefits, and disability benefits. Therefore, at the extensive margin we define individuals as working in a specific month when their main source of income stems from work, and as retired when their main source of income comes from pension benefits. Appendix A.1 shows how our labor market outcomes behave 3 years before and after pension eligibility. They show clear drops in labor force participation when an individual reaches the pension age. At the intensive margin we use the number of hours worked per month. We exclude self-employed individuals since exact workplace networks are unobserved within the data. Other difficulties with self-employed individuals include defining retirement as these workers often switch regularly between self-employment and retirement, and that we are unable to observe hours worked for self-employed individuals.[2]

We fix an individuals network at the age of 59.[3] This helps us to ensure that peers who

[2]We perform a robustness test with self-employed individuals included in the sample. We estimate the effect for the extensive margin outcomes and find similar effects. See section 6.4 for details.

[3]Since individuals might switch between jobs after the age of 59 peer groups can differ. In section 6.4 we

leave the workplace due to retirement are still considered as peers. Consequently, we remove all individuals younger than 59 since there is no peer group defined yet. After an individual reaches the age of 59 we track them throughout our entire sample period. Because we fix the network at 59, all peers in a network should be observable during the sample period. This is, however, not always the case since peers might pass away, emigrate, or simply not be observable in the data anymore.[4] The data allows to only identify deceased peers, meaning that we treat all other missing peers as missing observations.

We define pension eligibility as an indicator equal to one once an individual reaches the statutory pension age and zero otherwise. For each focal individual we collect the share of peers eligible for retirement, and the share of peers retired. We calculate these shares as the number of observable peers, retired or eligible, and divide this by the number of alive peers. Note that we keep the potentially missing peers in the denominator while these cannot enter either numerator. As a result, both shares mechanically decline when peer observability falls. We account for this variation by controlling for peer coverage which is defined as the share of still observable peers compared to the number of peers at the first month in the year an individual becomes 59. To reduce the effect of noisy observations we remove individuals with a peer coverage lower than 50%.[5]

4.3 Peer group construction

To construct our peer groups we utilize the novel person network data of Statistics Netherlands.[6] The data contains an individual and al its unique peers as links. Networks can overlap, meaning that two individuals can have similar peers, however, the individual itself is always excluded from its own network. We restrict the peers to be born between 1944 and 1962. This allows us to select for each individual, peers who are at most 6 years older

present a robustness test where we include only workers who do not switch between jobs after the age of 59.

[4]It could also simply be to reporting mistakes in the data

[5]We test the stability of our results by assessing the whether a different minimum peer coverage affects the results. See section 6.4 for the details.

[6]See here for the source file: (van der Laan, 2022)

or 3 years younger. This ensures the peers are in the same life cycle phase as the focal individual and are making decisions regarding retirement. Moreover, this increases the chance individuals actually interacting within the workplace.

Workplace peers Statistics Netherlands collects and links all individuals with an employment contract at the same company. They proceed in their derivation of colleagues as follows, within each month they select a maximum of 100 individuals within the same firm who are geographically closest in their residential address. If a firm has less than 100 employees, all employees are considered as colleagues. Subsequently, they sum all established colleagues for each month and again select a maximum of 100 colleagues who are geographically closest in their residential address. As individuals might change company within a year this ensures colleagues from both companies to be present in the set of colleagues. However, we do register the company and collective labor agreement for an individual in the first month of the year they turn 59, which means that the set of colleagues might contain individuals who are not from the company we assign to the focal individuals. We restrict our sample to companies with less than 100 employees. This ensures that a network will contain all colleagues and will increase the chance of coworkers interacting in the workplace. Of course, we are not sure these colleagues interact with one another, however, they serve as a potential reservoir of workplace peers.

Family peers This network layer contains all family relations for registered individuals on the first of January in the year they reach the age of 59. We select as family peers of the focal individual, the siblings, step-siblings, and potential partners of siblings. We restrict peers to this group as siblings are more likely to be in the same life-cycle and are more likely to interact than more distant family relations are. We exclude partners from the analysis as it is arguable whether these should be considered as peers. Moreover, (Rutten et al., 2023) study these exact effects of older partners on the retirement behavior of the younger partner. As with workplace peers we cannot be sure our selected peers interact, however they serve

as a potential group of family peers.

4.4 Descriptive statistics

Table 1: Descriptive statistics at age 59

	Workplace sample		Family sample	
	Mean	SD	Mean	SD
Observations	262,533		616,534	
Focal individual characteristics				
Female (%)	39.0	48.8	44.6	49.7
Migration background (%)	10.4	30.6	8.9	28.5
Single (%)	17.3	37.8	17.6	38.1
Couple (%)	82.3	38.2	81.8	38.6
Income at age 59 (€1,000)	42.1	45.5	44.9	39.2
Labor market outcomes				
Working (%)	93.1	25.4	95.0	21.8
Retired (%)	2.1	14.4	1.6	12.6
Monthly hours worked	133.0	55.2	132.0	49.3
Monthly hours worked, if positive	137.3	50.4	135.1	45.5
Peer-group characteristics				
Number of peers	7.6	7.4	2.7	1.8
Mean peer age	59.3	1.7	59.5	2.3
Variance of peer age	4.5	3.7	3.2	4.0
Share male peers (%)	62.0	37.8	53.7	34.3
Share retired peers (%)	6.0	15.6	14.2	27.0
Share AOW-eligible peers (%)	0.5	4.5	0.9	7.4
Mean peer hours	129.6	42.4	96.0	55.2
Mean peer wage (€1,000)	41.9	29.4	40.3	28.1

Notes: Statistics are measured at age 59. Percentages are on a 0–100 scale, corresponding standard deviations are multiplied by 100, and are expressed in percentage points. Income and wage variables are in thousands of euros. Peer-group variables are based on fixed age-59 networks.

Table 1 shows descriptive statistics for our workplace and family network specific samples at the baseline age 59. We note three main aspects of our data. In terms of demographics the samples are broadly comparable except for a slightly larger share of females in the family sample. Second, labor market outcomes show both network specific samples to be still very much attached to the labor market at age 59. Third, the network specific samples differ most in peer group characteristics. These differences are mainly a result of sample construction choices and the difference in sizes between the peer groups, and therefore, justify our choice to estimate to network specific models for the respective samples. Appendix A.2 plots the distribution for the network specific peer group sizes. It shows that within the family network the bulk of the peer groups contain 3 or less peers whereas in the workplace substantial amounts of larger peer groups are observed as well.

5 Empirical strategy

5.1 Identification problem

Our goal is to estimate the effect of peer retirement on focal individual labor supply. In the terminology of Manski (1993), this is an endogenous social effect: the effect of peers' behavior on individual behavior. A regression of peer retirement on individual labor supply does not generally recover this effect since peers are not randomly assigned.

Accordingly, we identify three main threats for identification: First, there are *contextual effects*, which mean that the focal individual responds to specific characteristics of their peers. For example, older peer groups may affect focal retirement behavior because they reveal a different workplace age structure, not because peers retire. Second, *correlated effects* refer to how an individual and their peers behavior is shaped by facing the same institutional environment such as the same HR policies. Moreover, it captures the process through which individuals select into particular peer groups; for instance, an individual may choose to work at a specific firm because they prefer its work environment. Third, peer retirement and focal

retirement may be jointly determined within the network, generating the reflection problem.

Concluding, a naive regression of peer retirement on individuals their labor market behavior cannot identify the endogenous social effects. Therefore, we utilize variation in peer retirement not driven by shared characteristics, common shocks, or simultaneous influence. This motivates our design of using variation in peers reaching the statutory pension age induced by a cohort-specific statutory retirement age schedule.

5.2 Instrumental variable design

Different statutory pension ages for peers in an individual's network creates policy-induced variation in the pension eligibility of peers dependent on their birth cohort. Therefore, two otherwise similar individuals can be exposed to different levels of retirement eligibility in their fixed networks. We use this variation as an instrument for peer retirement. Given this variation in statutory pension ages dependent on birth cohort we define the following indicator for peer j: $\text{Eligible}_{jt} = \mathbb{1}\{Age_{jt} \geq SPA_j\}$, where an individual is eligible if he reaches the age from which governmental pension benefits can be claimed. We define the network type $n \in W, F$ where W represents the workplace network and F the family network and use this to define N_i^n as the fixed age-59 peer group of focal individual i in network n. Next we define our endogenous variable, peer retirement, as the share of retired peers in the focal individuals' network:[7]

$$P_{it}^{ret,n} = \frac{1}{|N_i^n|} \sum_{j \in N_i^n, j \neq i} \text{Retired}_{jt} \tag{1}$$

Subsequently, we follow the methodology used in Oral et al. (2026) and define the instrument as the share of peers (indicated by j) of individual i that are eligible to claim their pension:[8]

[7] Note that here the share is calculated as the share of all observable peers divided by the number of alive peers as explained in section 4

[8] See footnote 8

$$Z_{it}^{n} = \frac{1}{|N_i^n|} \sum_{j \in N_i^n, j \neq i} \text{Eligible}_{jt} \tag{2}$$

We use this instrument to estimate the first-stage relationship which captures the extent to which peer AOW eligibility induces actual peer retirement. The second stage then uses this eligibility-induced component of peer retirement to estimate the effect on focal individual labor supply.

5.3 Baseline specification

We estimate the following two stage least squares specification for both the family and workplace networks. The first stage equation looks as follows:

$$P_{it}^{ret,n} = \alpha_l + \gamma_t + \zeta_b * Female_i + \pi Z_{it}^{n} + \rho X_{it}' + \theta \text{Eligible}_{it} + \epsilon_{it}. \tag{3}$$

And the second stage equation:

$$Y_{it} = \sigma_l + \delta_t + \eta_b * Female_i + \beta \hat{P}_{it}^{\text{ret}} + \phi X_{it}' + \xi \text{Eligible}_{it} + \nu_{it}, \tag{4}$$

where $P_{it}^{ret,n}$ represents the retirement behavior of peers or more specifically the average share of individual i's peers retired at time t.Z_{it}^{n} is our instrument defined in equation (2). Our outcome Y_{it} measures the labor force participation of the focal individual and can either be working hours, probability of working, or the probability of being retired.

Moreover, for the workplace sample we include collective labor agreement (CLA), gender by birth-month-year, and calendar month-year fixed effects. CLA fixed effects absorb differences between different CLA's which might differ by institutional retirement rules, retirement norms, or working conditions. Gender by birth-month-year fixed effects absorb all difference between genders and cohort specific retirement patterns. Calendar month-year fixed effects absorb all time varying differences in retirement patterns and institutions. In

the family sample we include both gender by birth-month-year, and calendar month-year fixed effects.

The vector X'_{it} contains the set of control variables. We control for the pension eligibility of the focal individual to exclude the direct effect of the retirement age on the labor force participation.[9] Other controls for the focal individual include: income percentile, migration background, and household type. Furthermore, we include peer group specific controls such as the average and variance of age, average income percentile, share of males, and the average hours worked by peers. On top of this, we control in the family network for whether the peer group contains a full sibling and in the workplace network for the share of peers with a migration background. Except for the age and eligibility controls we fix all controls at the age of 59 of the focal individual. Finally we control for the peer coverage as explained in section 4.2, and whether a peer passed away in the last 12 months.

Finally, β is our causal parameter of interest and measures the effect of an increase in peer retirement on focal individuals' labor force participation. Since, $P_{it}^{ret,n}$ is measured as a share from 0 to 1, we can interpret 0.1β as the effect of a 10 percentage points increase in the share of retired peers. Moreover, β measures the effect of peer retirement driven by peers reaching the statutory pension age.

5.4 Identifying variation & assumptions

Having explained the model, we now turn to what identifies exactly our causal effect of interest. Our identification comes from the variation of how much the focal individuals are exposed to peer retirement conditional on the focal individuals' own pension eligibility, focal gender by birth-month cohort, calendar month, CLA, and a set of individual and peer-group controls. This means we compare individuals with the same pension eligibility status within the same calendar month, same focal gender birth-month, and same CLA.

[9]The idea behind this is that if the instrument is valid for average peer outcomes, the individual level instrument must be valid for the focal individual as well meaning that it might affect the individual outcome. Therefore, failing to control for focals' individual pension eligibility leads to inconsistent estimates (von Hinke, Leckie, and Nicoletti, 2019).

Our approach rests on two main assumptions. First, the relevance condition requires a large and significant π in equation (3) meaning that the share of peers retired should be strongly driven by the share of peers eligible for pension. Second, the exclusion restriction must hold, which means that conditional on controls and fixed effects the pension eligibility of peers affects the labor force participation of the focal individual only through peer retirement. Several aspects of our empirical strategy support the plausibility of the exclusion restriction. We control for peer composition, focal individual pension eligibility, and include gender birth-month-year fixed effects. Together these deal with the age cohort of peers possibly affecting the labor force participation of the focal individual due to large correlations between the age of a focal individual and its peers. Moreover, our CLA and calendar month fixed effects control for CLA's or firms consisting of older peer groups which might reflect differences in retirement preferences, or institutional rules affecting the focal individual as well. Finally, we fix the peer group at age 59, at least 6 years before the SPA which makes it less likely that the observed peer composition reflects sorting in anticipation of their own retirement.

5.5 Peer similarity

When estimating peer effects the question may arise which peers are actually most influential, and do the effects differ by peer characteristics. To test this we split an individuals' peer group into similar and other peers. Types of similarity include, being the same gender, being in the same income decile, have a similar household type. If you include sector similarity mention it here!! Using these separated peer groups we construct separate peer-retirement, and peer-eligibility shares for each group. Both shares are defined relative to the full peer group size, meaning the two subgroup shares add up to one. If one subgroup contains no individuals this share is set as 0.

Because we split the peer group into two subgroups we now have two endogenous outcomes of interest: $P_{it}^{ret,n,\text{sim}}$ and $P_{it}^{ret,n,\text{oth}}$, which we both instrument by the share of eligible peers within its own subgroup. Furthermore, since pension eligibility in one subgroup is

correlated with eligibility in the other, we require both instruments, $Z_{it}^{n,\text{sim}}$ and $Z_{it}^{n,\text{oth}}$, to enter both first-stage equations:

$$P_{it}^{ret,n,\text{sim}} = \alpha_l + \gamma_t + \zeta_b Female_i + \pi_1 Z_{it}^{n,\text{sim}} + \pi_2 Z_{it}^{n,\text{oth}} + \rho X_{it}' + \theta \text{Eligible}_{it} + \epsilon_{it} \tag{5a}$$

$$P_{it}^{ret,n,\text{oth}} = \alpha_l + \gamma_t + \zeta_b Female_i + \kappa_1 Z_{it}^{n,\text{sim}} + \kappa_2 Z_{it}^{n,\text{oth}} + \rho X_{it}' + \theta \text{Eligible}_{it} + \epsilon_{it} \tag{5b}$$

The fitted values $\hat{P}_{it}^{ret,\text{sim}}$ and $\hat{P}_{it}^{ret,\text{oth}}$ from equations (5a) and (5b) then enter the second stage jointly:

$$Y_{it} = \sigma_l + \delta_t + \eta_b Female_i + \beta_1 \hat{P}_{it}^{ret,\text{sim}} + \beta_2 \hat{P}_{it}^{ret,\text{oth}} + \phi X_{it}' + \xi \text{Eligible}_{it} + \nu_{it} \tag{6}$$

Here, β_1 and β_2 can be interpreted as the effect of an increase in exposure of similar or other peers retiring, holding the retirement behavior of the other subgroup fixed. If more comparable peers exert greater influence, we expect $\beta_1 > \beta_2$; we formally test this with a Wald test of $H_0 : \beta_1 = \beta_2$.

5.6 Placebo test

To test whether our estimates capture the genuine peer effect we run a placebo test where we randomize the instrument pair over our sample of focal individuals. The idea is that assigning a random peer group to a focal individual should not lead to any peer induced changes in labor force participation for the focal individual. We approach the placebo test as follows: we randomize all peer variables within predefined randomization strata for each network specific sample. For strata we use year month and gender meaning that a focal individual can only receive the peer variables from a same gender individual within the same year month. As peer variables we include the endogenous share of retired peers, the share of eligible peers, and all peer group control variables discussed in section 5.3. This ensures the focal individual receives an actual but random peer group. We perform 250 permutations

and estimate our 2SLS model from equations (3) and (4) for all outcomes to obtain 250 point estimates for β. A successful placebo test means the distribution of placebo estimates $\{\hat{\beta}^{(p)}\}_{p=1}^{250}$ is centered on zero and that our true estimate $\hat{\beta}$ lies in its tail, formalized as a small placebo p-value:

$$p\text{-value} = \frac{1}{250}\sum_{p=1}^{250} \mathbb{1}\left\{\left|\hat{\beta}^{(p)}\right| \geq \left|\hat{\beta}\right|\right\} \tag{7}$$

A small p-value indicates our estimate reflects a genuine peer effect rather than random noise.

6 Results

In this section we present the results for the two stage least squares estimation discussed in the previous section. We start by discussing the baseline results for our main network samples, subsequently we present results on heterogeneity. Following this, we discuss results regarding whether similar peers exert stronger effects. Finally we present robustness tests, and a placebo test.

6.1 Baseline peer effects

Peer pension eligibility strongly predicts peer retirement behavior. Table 2 shows all results from our two stage least square estimation approach. Panel B displays the first stage results and shows strong and highly significant effect for the share of AOW-eligible peers on the share of retired peers, a 10 percentage point increase in the share of eligible peers leads to a 0.606 (workplace) or 0.595 (family) percentage point increase in the share of peers retired.[10] Furthermore, both network samples have large F-statistics of 144034 for the workplace and 608447 for the family network, where the larger statistic for the family

[10]We scale our estimates from table 2 by a factor 10 to interpret it as the effect of a 10 percentage points increase in peer retirement. Given the heterogeneity in a peer group it is more likely peers are retiring not all at once but rather in shares.

sample is likely driven by the sample size instead of instrument strength. The size of the F-statistics together with the significance and size of the coefficients leaves no doubt about the relevance for our instrument.

Exposure to more retired peers reduces the number of monthly working hours. Panel A displays the second stage results for all outcome variables. The peer effects on the monthly working hours of the focal individual are highly significant in both network samples and show that a 10 percentage point increase in the share of peers retired translates to a decrease in hours worked per month by 0.5280 in the workplace and 0.5055 in the family sample. These estimates correspond to approximately 0.65% and 0.64% of mean monthly hours in the respective estimation samples.

Peer retirement also reduces the extensive margin labor force participation. A 10 percentage point increase in the share of retired peers leads to a 0.27 percentage points increase in the probability of being retired in the workplace network and a 0.31 percentage points increase in the family network sample. For the outcome working a 10 percentage point increase in the share of retired peers leads to a 0.33 percentage points decrease in the probability of being retired in the workplace network and a 0.34 percentage points decrease in the family network sample. The coefficients are highly significant across both extensive margin outcomes and network samples. The coefficient for own pension eligibility are highly significant and large in magnitude across all outcomes and samples revealing the importance of including the focal individual eligibility itself.

Table 2: Baseline IV estimates by network

	Workplace network			Family network		
	Hours	Retired	Working	Hours	Retired	Working
Panel A. Second stage						
Share retired peers (instrumented)	−5.280***	0.027***	−0.033***	−5.055***	0.031***	−0.034***
	(0.419)	(0.002)	(0.003)	(0.206)	(0.001)	(0.001)
Own pension eligibility	−50.367***	0.577***	−0.416***	−48.066***	0.522***	−0.385***
	(0.225)	(0.001)	(0.002)	(0.117)	(0.001)	(0.001)
Outcome mean	80.84	0.374	0.557	79.60	0.375	0.568
Observations		32,518,094			73,728,793	
Panel B. First stage						
Share AOW-eligible peers		0.606***			0.595***	
		(0.002)			(0.001)	
Kleibergen–Paap Wald F-statistic		144,034			608,447	
Panel C. Fixed effects and controls						
Gender × birth-month-year FE	✓	✓	✓	✓	✓	✓
Month-year FE	✓	✓	✓	✓	✓	✓
Collective-agreement FE	✓	✓	✓			
Controls	✓	✓	✓	✓	✓	✓

Notes: The table reports two-stage least squares estimates. The endogenous variable is the share of retired peers in the focal individual's network, instrumented by the share of peers who are eligible for pension. Outcome means are calculated over person-month observations in the respective samples. Hours are measured as working hours per month. Retired and working are binary variables indicating whether an individual's main source of income in a month comes from pension or work. Peer shares range from zero to one; coefficients therefore refer to a one-unit (100 percentage point) increase in the share. The effect of a 10 percentage point increase is one-tenth of the reported coefficient. Standard errors are in parentheses. Workplace standard errors are clustered at the firm level; family standard errors are clustered at the individual level.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.2 Heterogeneity in peer effects

In this section we discuss the heterogeneity results for the peer effects established in section 6.1. The question we try answer here is: are retirement peer effects different for

specific subgroups. As subgroups we look at gender, income level, and household composition for both network specific samples, and look at company size for the workplace sample. SHOULD WE DISCUSS HERE WHY WE ADOPT A SPLIT SAMPLE APPROACH Instead of a pooled iv model? (We do this mainly because the pooled IV becomes too messy with our model)

Gender. Gender differences in peer effects are limited, although females' working hours are less responsive to family peers. Figure 2 displays the plotted coefficients for all outcomes across gender specific sub-samples within the greater network specific samples. For the extensive margin outcomes the confidence intervals suggest no difference in the estimated peer effect between man and woman. For the outcome hours worked, in the workplace females seem to experience slightly higher peer effects although confidence intervals still overlap. In the family sample the coefficient plot suggests the effect of being exposed to more retired peers to be higher for man than for woman. Note here that the average working hours for females in the 3 years before their pension age is around 103 in the family sample and 97 in the workplace sample, both substantially less than male average working hours of about 147 and 148 respectively.

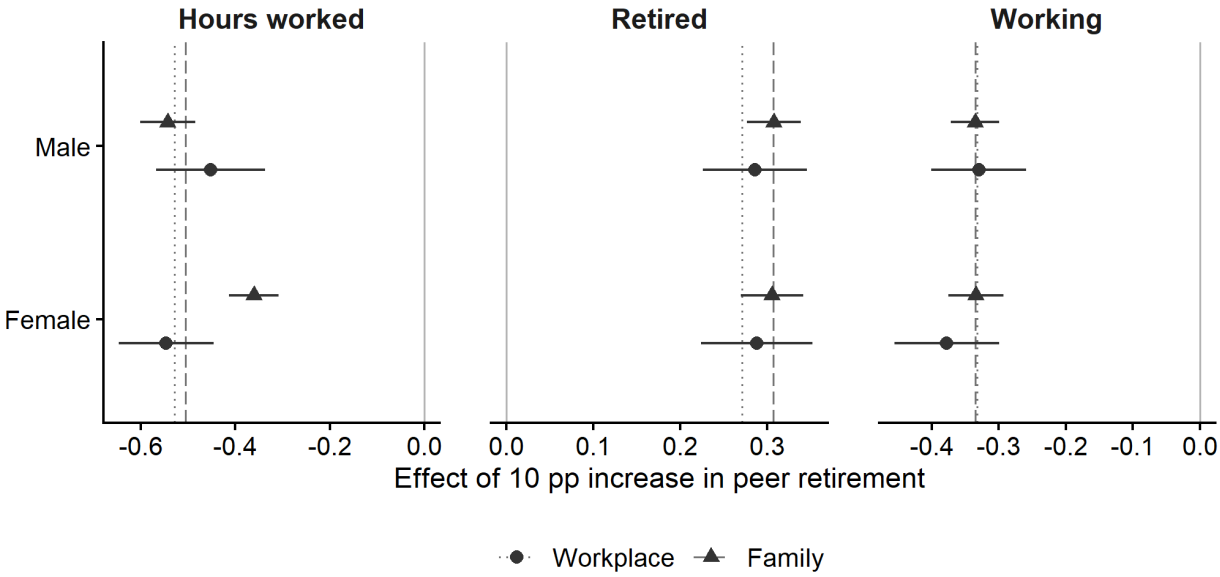


Figure 2: Heterogeneity by gender

Notes: The figure reports split-sample IV estimates by gender. Coefficients are scaled to reflect the effect of a 10 percentage point increase in the share of retired peers. Lines indicate9 5 percent confidence intervals. The estimates use the same fixed effects, controls, and clustering as the baseline specification, except that the gender split does not include gender interactions.

Income. Does income play a role for peer effects? Figure 3 shows the plotted coefficients for all outcomes across three income specific sub-samples within the greater network specific samples. For the family network sample the results suggest no heterogeneity in effect for different income levels. The results for the workplace sample suggest larger effects for individuals with high income and low income, with high income individuals having the largest effect. The effect for middle income individuals is not statistically significant different from zero but is substantially smaller than the effect for low and high income individuals.

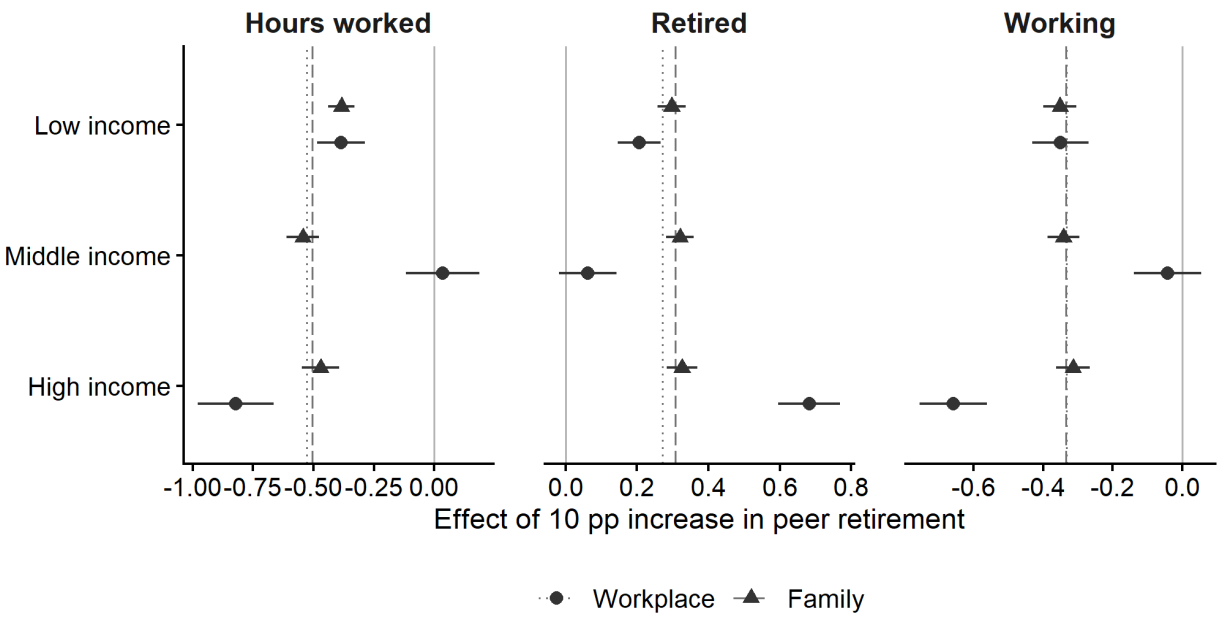


Figure 3: Heterogeneity by income

Notes: The figure reports split-sample IV estimates by income group. Coefficients are scaled to reflect the effect of a 10 percentage point increase in the share of retired peers. Whiskers indicate 95 percent confidence intervals. The estimates use the same fixed effects, controls, and clustering as the baseline specification.

Household composition. Single individuals are less responsive to their family peers' retirement behavior. Figure 4 shows the plotted coefficients for two types of household compositions. We divide the sample into individuals belonging to a single household or a household containing a couple. For the workplace network sample the results suggest no heterogeneity in effect between single households and couples. The results for the family sample suggest larger effects for individuals belonging to a couple living in a household.

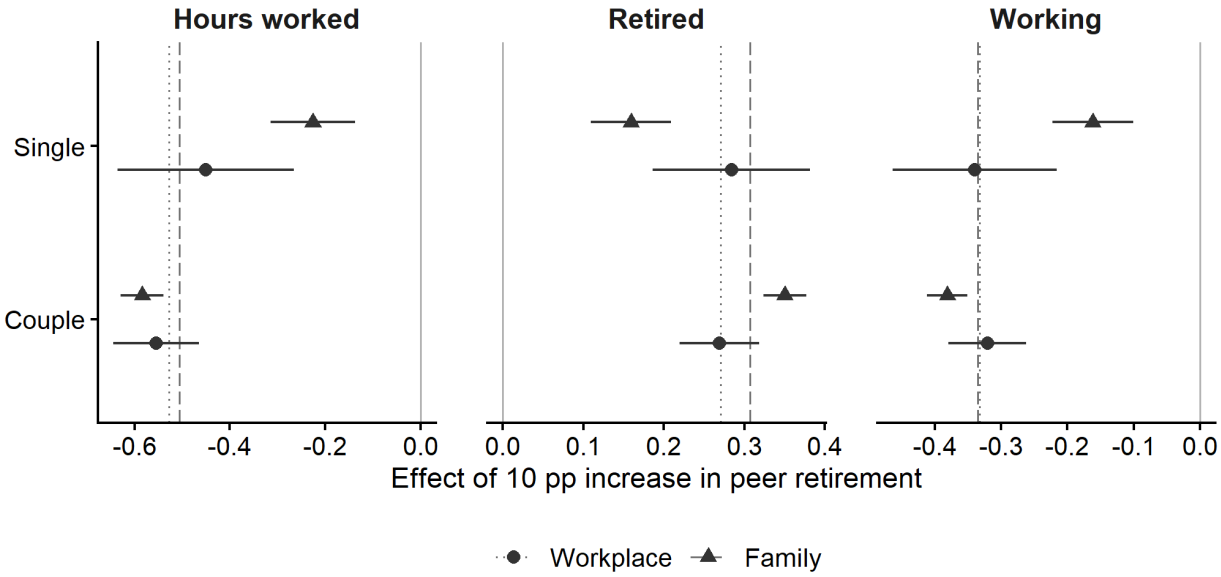


Figure 4: Heterogeneity by household composition

Notes: The figure reports split-sample IV estimates by household composition. Coefficients are scaled to reflect the effect of a 10 percentage point increase in the share of retired peers. Whiskers indicate 95 percent confidence intervals. The estimates use the same fixed effects, controls, and clustering as the baseline specification.

Firm size. Does having more colleagues imply larger peer effects or it means peer effects dissipating? Figure 5 below shows the coefficients for a heterogeneity analysis on whether the effect size differs by company size. We study this margin exclusively in the workplace sample as the company size is not likely to matter for family network effects. While all confidence intervals are somewhat overlapping the figure suggest slightly stronger effects for individuals working in a company with between 50 and 100 employees as compared to smaller companies.

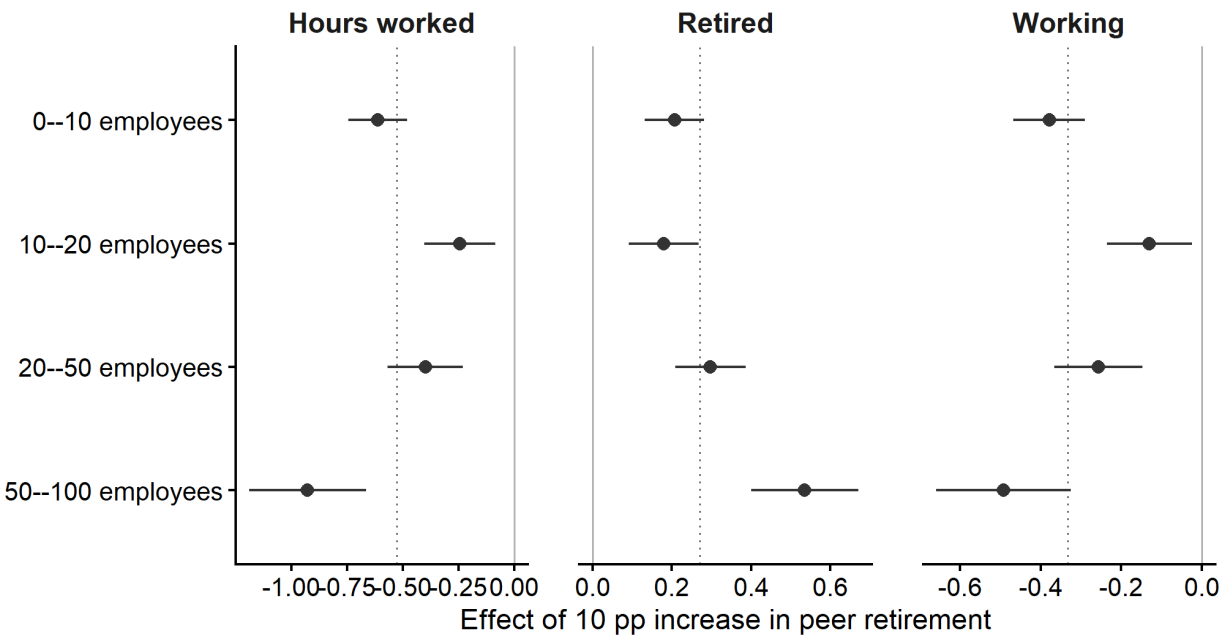


Figure 5: Heterogeneity by firm size

Notes: The figure reports split-sample IV estimates by firm size for the workplace network. Coefficients are scaled to reflect the effect of a 10 percentage point increase in the share of retired peers. Whiskers indicate 95 percent confidence intervals. The estimates use the same fixed effects, controls, and clustering as the baseline specification.

6.3 Peer similarity within networks

In this section we present second stage results on whether the peer effect is different for more similar peers. We show results for four margins of similarity, namely: gender, income level, household. Mention sector if we do include it The paired IV specifications rely on separate first stages for the same-peer and other-peer retirement shares. Appendix Tables 6-8 report the corresponding first-stage results. For both family and workplace specific networks and across all margins, the estimates show a clear structure: similar-peer eligibility primarily predicts same-peer retirement, while other-peer eligibility primarily predicts other-peer retirement.

Gender similarity. Workplace peer effects are driven mainly by peers from the same gender. Table 3 shows the estimation results for our paired instrument approach where we include the instrument for peers of the same gender, and peers of another gender. For the workplace specific sample, table 3 shows highly significant results for peers of the same gender as the focal individual. Scaling the coefficients by a factor 10 allows to interpret the estimates as follows: a 10 percentage point increase of peers of the same gender translates to a 0.8139 decrease in monthly hours worked, a 0.39 increase in the probability of being retired, and a 0.43 decrease in the probability of working. For peers of another gender the coefficient for hours and retired are insignificant or barely significant while the effect on working is significant. However, with p-values all below than 0.001 the results unambiguously show that individuals in the workplace respond stronger to peer retirement if these peers are of the same gender than of another.

Individuals do not respond differently to family peers from another gender. The results for the family specific network sample shows significant results for all outcomes both for peers of the same gender and peers of another gender. Focal individuals reduce their number of working hours per month by 0.5324 when they are exposed to a 10 percentage points increase of peers of the same gender retiring while this is a 0.478 decrease in hours for peers of another gender. This difference is statistically significant at a 5 % level (p-value of 0.002) and thus reveals some evidence that individuals respond stronger to family peers of the same gender. However, this difference is non-existent for the extensive margin outcomes (p-values of 0.644 and 0.313), where a 10 percentage points increase in retired peers of the same gender leads to an increase in the probability of being retired by 0.3 percentage point (0.033 reduction for the probability of working), and a 0.31 percentage points (0.34 reduction for the probability of working) increase induced by peers of another gender. Thus, for the extensive margin outcomes in the family specific network there is no evidence that individuals respond differently to peers of the same gender.

Table 3: Split-IV estimates by gender similarity

	Workplace network			Family network		
	Hours	Retired	Working	Hours	Retired	Working
Panel A. Second stage						
Same-gender peers	-8.139^{***}	0.039^{***}	-0.043^{***}	-5.324^{***}	0.030^{***}	-0.033^{***}
	(0.434)	(0.002)	(0.003)	(0.222)	(0.001)	(0.001)
Other-gender peers	0.089	0.005^{*}	-0.016^{***}	-4.780^{***}	0.031^{***}	-0.034^{***}
	(0.491)	(0.003)	(0.003)	(0.224)	(0.001)	(0.001)
Panel B. Equality test and sample						
p-value same = other	< 0.001	< 0.001	< 0.001	0.002	0.644	0.313
Observations		32,518,094			73,716,341	

Notes: Paired IV estimates; same/other-peer retirement shares (0–1) instrumented by same/other-peer AOW-eligibility shares. Hours in monthly hours; retired/working in probability units. SEs in parentheses. "p-value same = other" tests equality of the two coefficients. Workplace clustered by firm; family by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Income-decile similarity. Peer effects are stronger for peers with a similar income for monthly hours and the probability of working. Table 4 shows the results for the estimation where we try to find out whether individuals respond more strongly to peers with a similar income, which we measure by being in the same income decile. We find significant stronger effects for same income peers for the outcome working hours and being retired in the workplace (p-value ¡ 0.001), a 10 percentage points increase in the share of same-income peers being retired leads to a 0.7265 reduction in hours worked and an increase of 0.33 percentage points in the probability of being retired, whereas for other-income individuals the reduction in hours is 0.4014 per month and the probability of being retired is increased only by 0.24 percentage points. Thus, for these outcomes this suggests that individuals in the workplace respond more strongly to peers of the same income. On the other hand, the probability of working decreases by 0.33 percentage points as a response to both same-income peers and other-income peers increasing their retirement by 10 percentage points, meaning that the evidence does not suggest individuals to change their probability of working more strongly to same-income peers (p-value of 0.898).

In the family network, individuals their extensive margin labor force participation is more responsive to peers with a similar income. Table 4 shows no significant difference in peer effects for same-income peers as compared to other-income peers for the hours worked

(p-value of 0.809). Here, a 10 percentage points increase in the share of same-income peers retired translates to a 0.5088 reduction in monthly hours worked, and a 0.5044 decrease in the hours worked for other-income peers. For the extensive margin outcomes there is evidence for a smaller response to peers from the same income decile than to peers from another income decile. A 10 percentage points increase in the share of same-income peers retired leads to a 0.26 percentage points increase in the probability retired and a 0.29 percentage points decrease in the probability of working, whereas for other income peers this translates to a 0.33 percentage points increase of the probability of being retired and a 0.35 percentage points decrease in the probability of working (p-value ¡ 0.001 for both outcomes).

Table 4: Split-IV estimates by income-decile similarity

	Workplace network			Family network		
	Hours	Retired	Working	Hours	Retired	Working
Panel A. Second stage						
Same-income peers	-7.265^{***}	0.033^{***}	-0.033^{***}	-5.088^{***}	0.026^{***}	-0.029^{***}
	(0.473)	(0.003)	(0.003)	(0.246)	(0.001)	(0.002)
Other-income peers	-4.014^{***}	0.024^{***}	-0.033^{***}	-5.044^{***}	0.033^{***}	-0.035^{***}
	(0.443)	(0.002)	(0.003)	(0.212)	(0.001)	(0.001)
Panel B. Equality test and sample						
p-value same = other	< 0.001	< 0.001	0.898	0.809	< 0.001	< 0.001
Observations		32,518,094			73,716,341	

Notes: Paired IV estimates; same/other-peer retirement shares (0–1) instrumented by same/other-peer AOW-eligibility shares. Hours in monthly hours; retired/working in probability units. SEs in parentheses. "p-value same = other" tests equality of the two coefficients. Workplace clustered by firm; family by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Household-type similarity. Peers from another type of household have a greater influence on monthly hours and retirement whereas peers from the same household exert a large influence on the probability of working. Table 5 displays the results for the paired 2SLS approach for same and other-household peers which reveal whether the focal individual responds more strongly to peers coming from the same type of household. The intensive margin outcome for the workplace network shows that a ten percentage points increase in the share of same-household peers retired translates to a 0.4690 reduction in working hours per month whereas this translates to a 0.6815 reduction in working hours for peers from

another household. This difference is statistically significant (p-value ¡ 0.001) meaning that the evidence indicates that individuals in the workplace their working hours respond more strongly to peers from the another household type. For the extensive margin outcomes a 10 percentage points increase in the share of same-household type peers retired leads to an increase of 0.26 percentage points in the probability of being retired and a 0.35 percentage points decrease in the probability of working. For other-household type peers a 10 percentage points increase leads to a 0.32 percentage points increase in the probability of being retired and a 0.27 percentage points decrease in the probability of working. For both extensive margin outcomes the difference between the effect for same and other-household type peers is statistically significant at a 1% level (p-value of 0.003 and 0.002 respectively), however, for the probability of being retired individuals are more affected by peers from another household whereas for the probability of working peers are more affected by peers from the same household type.

Within the family, individuals their extensive margin labor force participation responds more strongly to peers with a the same type of household. The right side of table 5 displays the results for the family specific network sample. The first column shows there is no significant difference between how an individual adjusts its working hours to same or other-household peers (p-value of 0.302). For a 10 percentage points increase in the share of same-household type peers (other household type peers), the focal individual decreases its working hours by 0.5011 (0.5194). The extensive margin outcomes do show a statistically significant difference between focal individual responses to same or other-household type peers (p-values < 0.001), namely individuals respond more strongly to peers from the same household-type. A ten percentage points increase in the share of peers retired from the same household type translates to a 0.33 percentage points increase in the probability of being retired and a 0.37 decrease in the probability of working, whereas, for peers from another household type this response is a 0.26 percentage points increase in the probability of being retired and a 0.26 percentage points decrease in the probability of working.

Table 5: Split-IV estimates by household-type similarity

	Workplace network			Family network		
	Hours	Retired	Working	Hours	Retired	Working
Panel A. Second stage						
Same-household type peers	-4.690^{***}	0.026^{***}	-0.035^{***}	-5.011^{***}	0.033^{***}	-0.037^{***}
	(0.438)	(0.002)	(0.003)	(0.213)	(0.001)	(0.001)
Other-household type peers	-6.815^{***}	0.032^{***}	-0.027^{***}	-5.194^{***}	0.026^{***}	-0.026^{***}
	(0.499)	(0.003)	(0.003)	(0.238)	(0.001)	(0.002)
Panel B. Equality test and sample						
p-value same = other	< 0.001	0.003	0.002	0.302	< 0.001	< 0.001
Observations		32,518,094			73,716,341	

Notes: Paired IV estimates; same/other-peer retirement shares (0–1) instrumented by same/other-peer AOW-eligibility shares. Hours in monthly hours; retired/working in probability units. SEs in parentheses. "p-value same = other" tests equality of the two coefficients. Workplace clustered by firm; family by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.4 Robustness and validation

To show our results discussed in the previous section hold up if we adjust the assumptions or data choices made, we present a number of robustness and validation tests.

Alternative specifications and sample restrictions Our results are robust to multiple alternative specifications and data choices. Table 9 in appendix B.2, describes multiple robustness checks for our baseline 2SLS model for the workplace specific network sample. Our baseline results are robust to using alternative clusters for our standard errors, and to using sector or firm fixed effects instead of collective labor agreement fixed effects. Moreover, our results hold up if we relax or tighten our sample restriction regarding peer group coverage, include individuals working in firms with more than 100 employees, and if we include self-employed individuals in the sample. Finally, including only individuals for whom we can observe that they do not switch jobs does not significantly change our results either.

Moving on to the family specific network sample, table 10 in appendix B.2 displays robustness checks for the baseline estimates for this sample. The table shows our results are stable to adding collective labor agreement fixed effects. Furthermore, adapting a alternative threshold for peer coverage does not change our results neither.

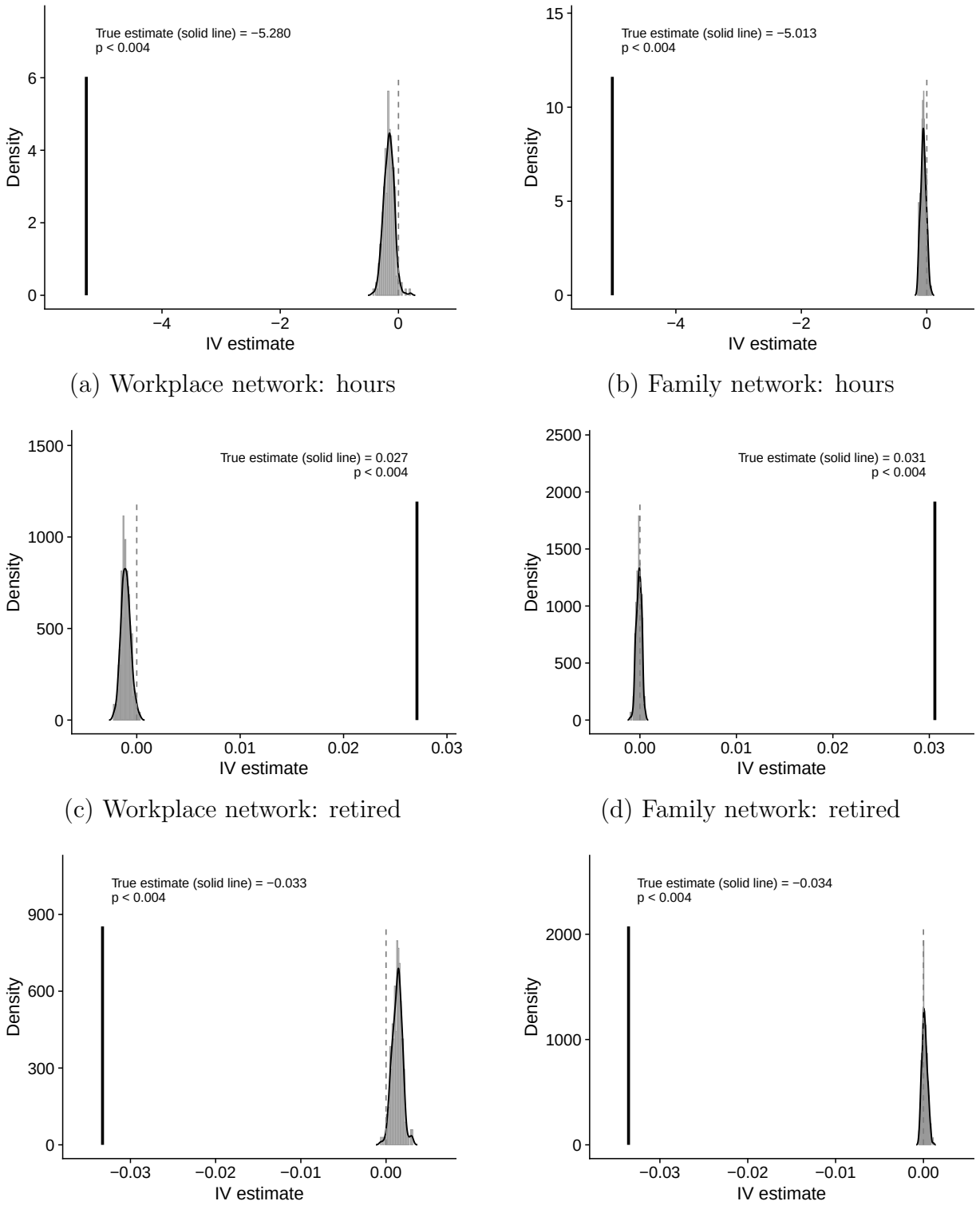
Placebo networks Peer effects drop drastically when randomizing peers over our sample. Figure 6 shows the distribution of the 250 placebo estimates, the true estimate from table 2 is indicated by a solid line. Figure (a) and (b) display the placebo test results for the intensive margin outcomes. Both figures show the true estimates lying well beyond the range of the placebo estimates, indicated as well by the p-values smaller than 0.004. Panels (c–f) tell the same story, none of the placebo distributions lie anywhere near our model estimates (p-values < 0.004). Hence, these results show our estimates for peer effects can not be reproduced using random peer groups. Note however, that none of the placebo distributions are exactly centered around zero, indicating that the randomized networks seem to generate small mechanical variation producing these non-zero results. See figure 10 in appendix B.3 for a more detailed figure displaying the distribution of the placebo estimates.

7 Discussion

7.1 Magnitude of peer effects

Our baseline estimates suggest statistically significant peer effects, begging the question, how large are the peer effects on an economic scale? We find positive peer effects, and scaling our estimates we find that a 10 percentage points increase in the share of peers retired results in a reduction of working hours by roughly half an hour per month, which translates to roughly 0.65% of the average monthly hours. Moreover, a 10 percentage points increase in peer retirement increases the probability of retirement by about 0.3 percentage points. Although, the magnitude of our estimates might not seem large, they should be contextualized within the situation where peer retirement exposure changes simultaneously for many focal individuals when a specific cohort reaches pension eligibility. Therefore, the peer effects should be interpreted as identifying the indirect effect beyond the direct effect of an increase in the statutory pension age.

Besides, peer effects gain economic significance when we interpret them as the effect of one



(a) Workplace network: hours

(b) Family network: hours

(c) Workplace network: retired

(d) Family network: retired

(e) Workplace network: working

(f) Family network: working

Figure 6: Placebo estimates from randomized peer networks

Notes: The figure shows the distribution of IV estimates obtained from randomized peer networks. The dashed vertical line marks zero. The arrow and label at the panel boundary indicate the true model estimate from the observed network when the observed estimate lies outside the displayed range. Peer-retirement shares range from zero to one.

additional peer retiring. Up until now we interpreted the peer effect as the change in labor force participation to a 10 percentage points increase in peer retirement. Using the average size of peer groups we can roughly calculate the peer effect peer one additional retired peer. The average peer group size in the workplace is 7.6 meaning that one additional peer retired corresponds to around 13 percentage points. Translating this to our estimates and outcomes implies that one additional retired peer decreases the monthly hours by 0.7 and increases the probability of retirement by 0.35 percentage points. The average peer group size in the family network sample is 2.7 which allows us to approximate the percentage change of one extra peer retiring by 37 percentage points. Multiplying this with our baseline estimates then gives, a 1.87 monthly hour reduction and a 1.15 percentage points increase in retirement for each additional peer retiring. These calculations suggest that the per additional retired peer effect is substantially larger for the family network than for the workplace network. However, note that peer group sizes differ substantially and above calculations should be interpreted merely as illustrative.

Finally, since we include focal individuals' own pension eligibility in the model we can discuss the direct effect of an increase in the pension age. The corresponding estimates show that focal individuals' pension eligibility decreases monthly hours by roughly 50 hours and increase the probability of retiring by more than 50 percentage points. The direct effect of the pension reform, estimated by the effect of focal individuals' own pension eligibility are closely comparable to previous evidence on the effects of statutory pension age reforms in the Netherlands. For example, Rabaté et al. (2024) find that an increase in the statutory pension age reduces retirement by 59 percentage points. Although the estimates are not directly comparable, the magnitude is close to our baseline estimates which show an increase of more than 50 percentage points.

7.2 Comparison with related causal evidence

While it is difficult to directly compare our results with the literature we try to contextualize our estimates within the literature of causal evidence of retirement peer effects. We find broadly comparable estimates in terms of direction of causality but smaller magnitudes. However, care is required when comparing specific magnitudes as differences in sample, network definitions, could obscure the comparison.

Two closely related papers, (Badalyan, 2025; Sturm, 2025), find positive retirement peer effects for woman in the workplace exploiting the abolishment of an early retirement reform in Germany. Sturm (2025) finds that a 10 percentage points increase in the share of workplace peers born after reform increases employment probabilities by 3.3 to 3.5 percentage points while Badalyan (2025) find a 1.33 percentage points increase for a 10 percentage points increase in peers retiring in response to the reform. These estimates are substantially larger than our estimate of a 0.33 percentage points decrease in employment for a 10 percentage points increase in the share of peers retired. However, both papers study a specific salient early retirement reform for more narrowly defined peer group. Moreover, we study a broader monthly outcome over multiple ages whereas they focus particularly on employment at or around the age of 62. Finally, we find suggestive evidence of larger workplace peer effects for woman coinciding with these papers focusing on peer effects for woman.

The most important comparison for our paper is with the paper by Oral et al. (2026) who study the retirement peer effects of the abolishment of an early retirement arrangement in the Netherlands in 2006. Using a much broader definition for workplace peers they find a spillover effect of about 1.3 months of delayed retirement for every year that a workplace peer delays retirement but they also find a 5.9 months months delay in retirement for very similar coworkers. For siblings, they find insignificant results for all siblings but for similar siblings they find an effect of 10.1 months.

Although our outcome is different we can, compare their estimates with ours by interpreting the probability of being retired in month t as a cumulative distribution of retirement

timing. This assumption requires retirement to be an absorbing state, which in theory is not the case for our outcome, however, for more than 75% of our sample this is the case meaning for the sake of comparison we assume it to be a completely absorbing state. Under above assumptions and approximations, we can interpret β as capturing the extent to which a shift in the average retirement timing of a focal individuals' peers translates into a shift in the individuals' own retirement timing.[11] This means we can interpret it as the area under the change in the retirement status curve over 12 months. Subsequently, we can apply this interpretation to our own estimate of $\beta = 0.027$ for the workplace and $\beta = 0.031$ for the family sample, meaning that a one year shift in average peer retirement timing corresponds to an approximate shift in own retirement timing of:

$$0.027 \times 12 = 0.324 \text{ months.}$$

For the family network, the corresponding estimate is

$$0.031 \times 12 = 0.372 \text{ months.}$$

This means our workplace estimate is around 4 times smaller than the colleague peer effect found by Oral et al. (2026). Their estimates for similar coworkers and similar siblings are considerably larger. This difference can be explained by multiple factors. First, Oral et al. (2026) study a one time pension reform making early retirement substantially less attractive for a large share of their sample. We us gradual increase in the statutory pension age meaning that the variation in peer group retirement is likely to be smaller than with the one time large early retirement reform. Hence, we rely on less identifying variation which can explain the smaller magnitudes of our estimates. Moreover, we rely on a more strict definition of peer networks, and use different outcomes. Finally, it is very likely that the group of compliers differs substantially, thereby affecting the estimates as well.

[11] See appendix B.4 for the formal derivation

7.3 Potential mechanisms & policy implications

We do not specifically test for the underlying mechanisms of our results. Nevertheless our heterogeneity and peer similarity results are informative for which mechanisms might play a role. Previous literature mainly discuss social norms/conformity, information transmission and complementarities as possible mechanisms (Sturm, 2025; Badalyan, 2025; Oral et al., 2026). Two main mechanisms for peer effects could be a social norm mechanism or an information story. With the former the idea is that individuals adjust their behavior to fit the norm in their peer peer group, in our case workplace or family. The latter, can be explained as peers sharing information on retirement possibilities within the company or simply informing peers on the perks of being retired. We cannot distinguish between these mechanisms however, in the workplace we find stronger effects for peers of the same gender which can be suggestive of both a homophily/social identify mechanism, individuals react stronger to peers which more similar to them, and a mechanism of information transmission.

Within workplace there is an interplay between the peer effect and the income effect on retirement. High income individuals are most responsive to an increase in peers retiring, they reduce their labor force participation more than middle and low income individuals. Peer effects are absent for middle income individuals. This seems to suggest that when individuals react to other peers retiring income can act as a enabling or limiting factor for how much to reduce their labor force participation.

Within the family network, a possible mechanism could be leisure complementarities and coordination which holds that family members might coordinate their retirement in order to enjoy their leisure together which increase the value of their leisure. Another mechanism could be that someone retires to become a caregiver for an earlier retired family member. The similar magnitude of family and workplace effects suggest that retirement spillovers are present both in the workplace and in the family where leisure complementarities and coordination provide some explanation.

Policy implications Peer spillovers amplify the direct effect of the pension reform by roughly 3% for retirement timing and 6% for working hours. Increasing the state pension age leads to indirect spillover effects further amplifying the direct effect of the pension reform. Consequently, individual level estimates of pension reforms may omit social spillovers. Using the approximate retirement timing interpretation from section 7.2, and a three months increase in the pension age,[12] we derive the per person spillover and direct effect as follows:

$$\underbrace{3 \times \text{own pension eligibility effect}}_{\text{direct reform effect}}, \qquad \underbrace{3 \times \text{first-stage effect} \times \text{peer effect}}_{\text{peer spillover effect}}.$$

Accordingly, a three months increase in the state pension age translates to a 0.0491 months delay in retirement for the workplace sample and a 0.0553 months delay in retirement for the family sample. For the intensive margin outcome working hours we obtain the per person effects similarly: here three months later eligibility leads to 9.6 additional working hours while for the family network, the corresponding value is 9.02 extra working hours per person. Using the per person spillover effects of a 3 month increase in the statutory pension age we can then express the peer spillover effect as a percentage amplification of the direct reform effect. This means that the the peer spillover in the workplace corresponds to approximately 2.8% of the direct retirement effect and in the family sample roughly to 3.5%. For working hours this amounts to 6.4% of the direct effect of the reform in the workplace and 6.3 % in the family sample.

The macro scale peer effects amount to roughly 1.6-1.8% of the maximum three-month delay and to an increase in the total working hours by 2.3-2.4%. The paragraph above calculated the per person social spillover effects of increases in the statutory pension age. In order to calculate the macro scale effects we perform a back of the envelope calculation and assume a population of 200,000 individuals making retirement decisions.[13] This allows us to

[12] We use a 3 month increase in the state pension age here as it is the most common increase within our sample, see table 1

[13] Our workplace sample contains roughly 260,000 individuals and the family sample around 616,000, see table 1. We use 200,000 as an illustrative population.

calculate the macro scale effects as the number of individuals times the per person peer effect. The macro workplace spillover effects of a three months increase in the statutory pension age, on retirement amount to 818 person years of delayed retirement and 921 person years for the family sample. To make this more intuitive we scale this number by the maximum additional person years which shows that the retirement spillover amounts to roughly 1.64% for the workplace and 1.84 % in the family sample.[14] Similarly, a 3 months increase in the state pension age leads to 1.92 million additional working hours through workplace peer effects and 1.804 million extra working hours through family network effects. Subsequently scaling this by the total working hours of our population gives that the peer effect corresponds to a 2.4 % increase in intensive margin labor supply in the workplace and 2.3% increase in the family sample.[15]

We provide illustrative macro scale estimates of the social spillover of increasing the pension age. Note that we do not sum the peer effects from the family and workplace sample together. Individuals can be exposed through both networks simultaneously, however, since we estimate them separately we cannot distinguish between the two channels. Moreover, we only estimate first round spillovers, i.e. our estimates do not reflect further network feedback which is likely to affect the spillover effects of the pension reform.

8 Conclusion

[14] The maximum person years can be calculated by multiplying our assumed population with three months: $200,000 \times 3/12 = 50,000$

[15] We calculate the total working hours by using the average working hours in our population from 1: $200,000 \times 3 \times 133(132) \approx 79.8(79.2)$ million hours in the workplace and family samples respectively.

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A Additional Descriptive Evidence

A.1 Visualization of outcome definitions

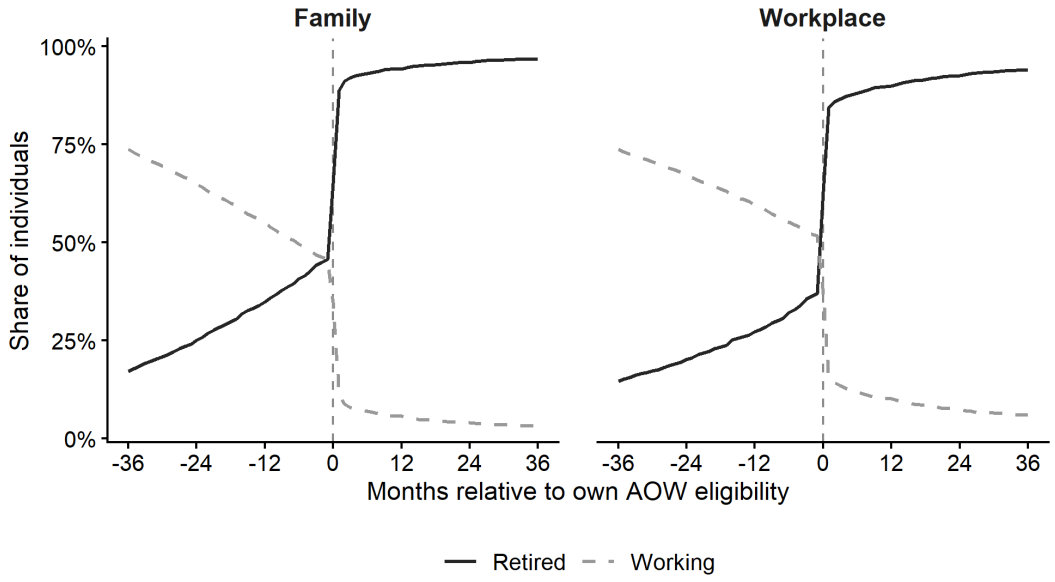


Figure 7: Labor-market outcomes around own AOW eligibility

Notes: The figure plots the share of individuals retired and working by months relative to own AOW eligibility. The vertical dashed line marks the month in which the focal individual reaches AOW eligibility. Outcomes are shown separately for the workplace and family analysis samples.

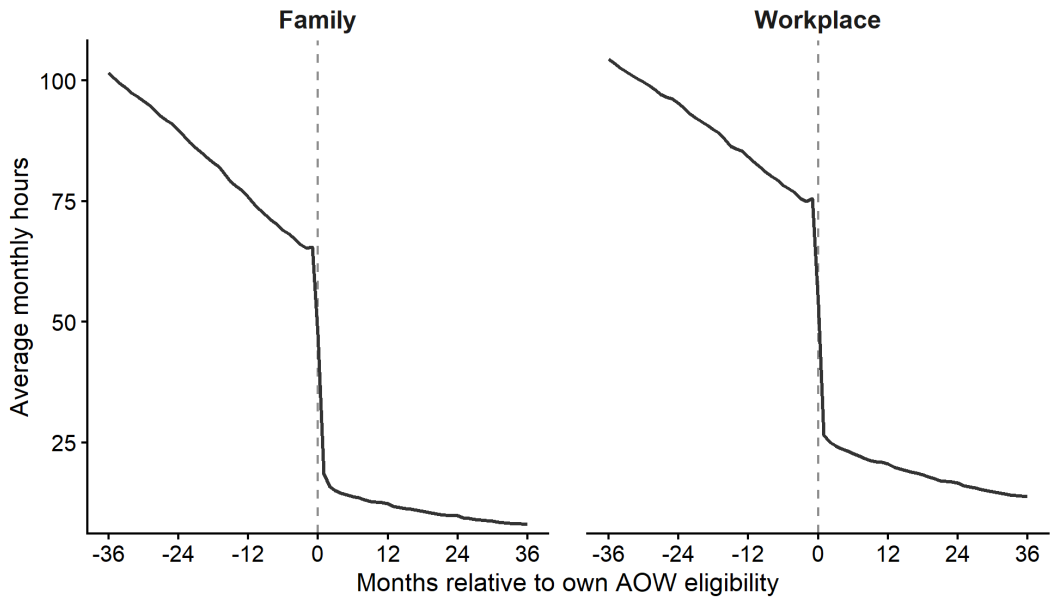


Figure 8: Monthly hours worked around own AOW eligibility

Notes: The figure plots average monthly hours worked and average monthly hours among workers by months relative to own AOW eligibility. The vertical dashed line marks the month in which the focal individual reaches AOW eligibility. Outcomes are shown separately for the workplace and family analysis samples.

A.2 Peer-Group Size Distributions

Figure 9 presents the distribution of peer-group sizes in the workplace and family network samples. Peer groups are generally smaller in the family network than in the workplace network.

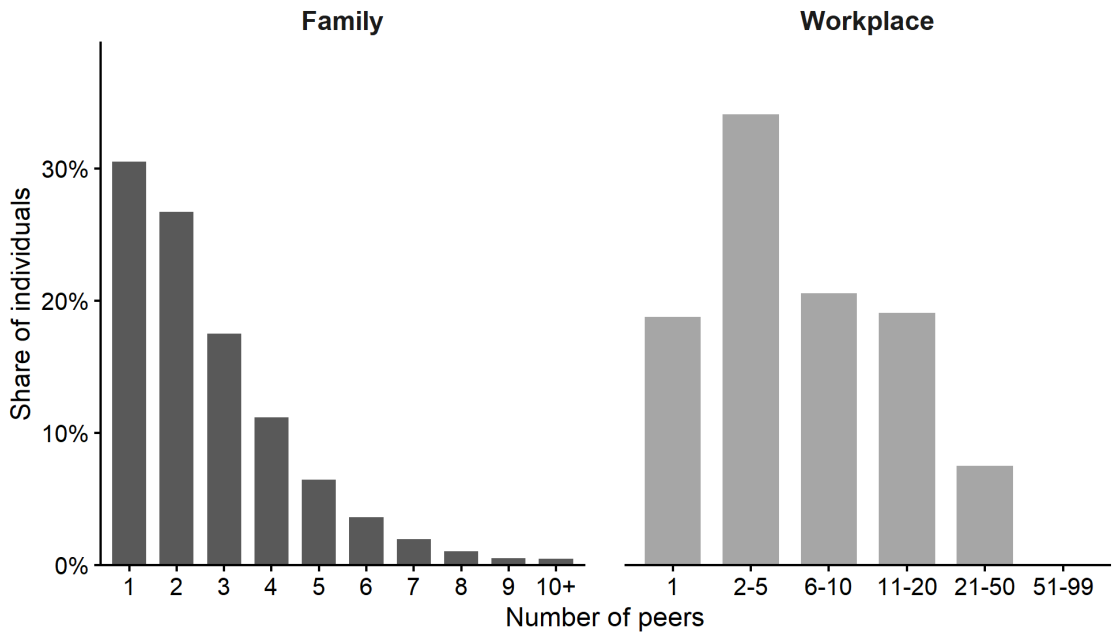


Figure 9: Distribution of Peer-Group Sizes by Network Type

Notes: The figure shows the distribution of the number of peers in the workplace and family network samples.

B Additional results

B.1 First-stage results for split-IV specifications

This subsection reports the first stage estimates of the paired 2SLS approach from 6.3. Since there are two instruments, one for similar peers and one for other peers, each table reports the first stage estimates for the two estimated first stage models. For all similarity margins the estimates show strong first stage relationships.

Gender

Table 6: First-stage estimates for split-IV specification: gender similarity

	Same-gender retirement	Other-gender retirement
Panel A. Workplace network		
Same-gender eligibility	0.7352^{***}	-0.1144^{***}
	(0.0015)	(0.0010)
Other-gender eligibility	-0.1995^{***}	0.7797^{***}
	(0.0013)	(0.0018)
Observations	32,518,094	
Panel B. Family network		
Same-gender eligibility	0.7711^{***}	-0.1755^{***}
	(0.0007)	(0.0006)
Other-gender eligibility	-0.1855^{***}	0.7791^{***}
	(0.0006)	(0.0007)
Observations	73,716,341	

Notes: The table reports first-stage estimates for the gender split-IV specification. The dependent variables are the same-gender and other-gender peer-retirement shares. Same-gender and other-gender peer-retirement shares are instrumented by the corresponding same-gender and other-gender shares of AOW-eligible peers. Peer-retirement and peer-eligibility shares range from zero to one. All specifications include the same controls and fixed effects as the corresponding second-stage specification. Workplace standard errors are clustered at the firm level; family standard errors are clustered at the individual level.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Income

Table 7: First-stage estimates for split-IV specification: income-decile similarity

	Same-income retirement	Other-income retirement
Panel A. Workplace network		
Same-income eligibility	0.8167***	−0.1991***
	(0.0015)	(0.0012)
Other-income eligibility	−0.1139***	0.7130***
	(0.0010)	(0.0015)
Observations	32,518,094	
Panel B. Family network		
Same-income eligibility	0.8620***	−0.2683***
	(0.0006)	(0.0007)
Other-income eligibility	−0.0998***	0.6947***
	(0.0005)	(0.0007)
Observations	73,716,341	

Notes: The table reports first-stage estimates for the income-decile split-IV specification. The dependent variables are the same-income-decile and other-income-decile peer-retirement shares. Same-income and other-income peer-retirement shares are instrumented by the corresponding same-income and other-income shares of AOW-eligible peers. Peer-retirement and peer-eligibility shares range from zero to one. All specifications include the same controls and fixed effects as the corresponding second-stage specification. Workplace standard errors are clustered at the firm level; family standard errors are clustered at the individual level.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Household

Table 8: First-stage estimates for split-IV specification: household-type similarity

	Same-household retirement	Other-household retirement
Panel A. Workplace network		
Same-household eligibility	0.6710^{***}	-0.0699^{***}
	(0.0016)	(0.0007)
Other-household eligibility	-0.2154^{***}	0.8462^{***}
	(0.0013)	(0.0015)
Observations	32,518,094	
Panel B. Family network		
Same-household eligibility	0.7027^{***}	-0.1075^{***}
	(0.0007)	(0.0005)
Other-household eligibility	-0.2448^{***}	0.8459^{***}
	(0.0007)	(0.0007)
Observations	73,716,341	

Notes: The table reports first-stage estimates for the household-type split-IV specification. The dependent variables are the same-household-type and other-household-type peer-retirement shares. Same-household and other-household peer-retirement shares are instrumented by the corresponding same-household and other-household shares of AOW-eligible peers. Peer-retirement and peer-eligibility shares range from zero to one. All specifications include the same controls and fixed effects as the corresponding second-stage specification. Workplace standard errors are clustered at the firm level; family standard errors are clustered at the individual level.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.2 Robustness of baseline IV estimates

Table 9: Robustness of baseline IV estimates: workplace network

	Hours	Retired	Working	Observations
Panel A. Alternative specifications				
Baseline	−5.280***	0.027***	−0.033***	32,518,094
	(0.419)	(0.002)	(0.003)	
Two-way clustering	−5.280***	0.027***	−0.033***	32,518,094
	(0.419)	(0.002)	(0.003)	
Sector FE instead of collective-agreement FE	−5.049***	0.026***	−0.032***	32,518,094
	(0.421)	(0.002)	(0.003)	
Firm FE instead of collective-agreement FE	−4.679***	0.028***	−0.031***	32,518,061
	(0.413)	(0.002)	(0.003)	
Panel B. Network construction and sample restrictions				
Coverage threshold: 25%	−5.195***	0.026***	−0.032***	32,773,602
	(0.419)	(0.002)	(0.003)	
Coverage threshold: 75%	−5.223***	0.029***	−0.035***	29,102,462
	(0.443)	(0.002)	(0.003)	
Including large firms	−5.618***	0.033***	−0.027***	128,184,059
	(0.925)	(0.005)	(0.006)	
Non-movers only	−4.759***	0.022***	−0.028***	19,057,170
	(0.541)	(0.003)	(0.004)	
Including self-employed	−4.641***	0.027***	−0.033***	36,016,883
	(0.414)	(0.002)	(0.003)	

Notes: The table reports robustness checks for the baseline workplace IV specification. All entries report the coefficient on the share of retired workplace peers. The endogenous variable is instrumented by the share of workplace peers who are eligible for pension. Peer-retirement shares range from zero to one. Hours are measured as working hours per month. Retired and working are binary variables indicating whether an individual's main source of income in a month comes from pension or work. Unless otherwise indicated, specifications include the baseline controls and fixed effects from Table 2. The baseline specification includes gender-by-birth-month-year fixed effects, month-year fixed effects, and collective-agreement fixed effects, with standard errors clustered at the firm level. Two-way clustering clusters standard errors at the firm and individual level. Standard errors are in parentheses.

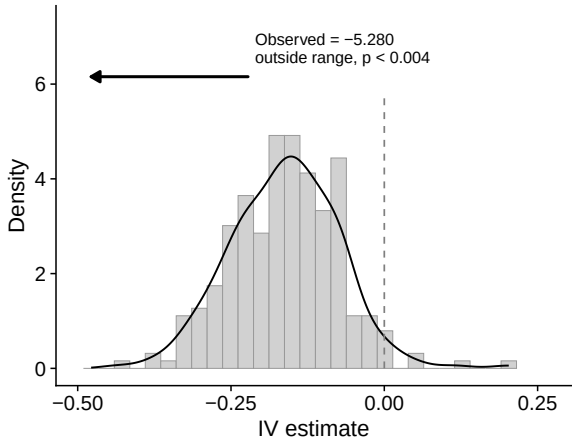
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Robustness of baseline IV estimates: family network

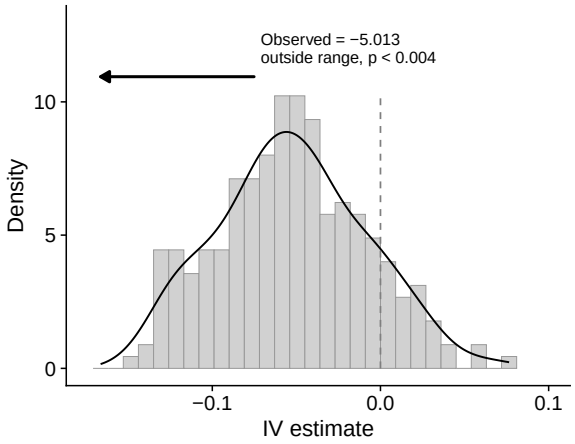
	Hours	Retired	Working	Observations
Panel A. Alternative specifications				
Baseline	-5.055^{***}	0.031^{***}	-0.034^{***}	73,728,793
	(0.206)	(0.001)	(0.001)	
Adding collective-agreement FE	-5.013^{***}	0.031^{***}	-0.034^{***}	73,728,793
	(0.205)	(0.001)	(0.001)	
Panel B. Network construction and sample restrictions				
Coverage threshold: 25%	-5.045^{***}	0.031^{***}	-0.033^{***}	74,069,905
	(0.206)	(0.001)	(0.001)	
Coverage threshold: 75%	-5.089^{***}	0.032^{***}	-0.035^{***}	62,798,515
	(0.218)	(0.001)	(0.001)	

Notes: The table reports robustness checks for the baseline family IV specification. All entries report the coefficient on the share of retired family peers. The endogenous variable is instrumented by the share of family peers who are eligible for pension. Peer-retirement shares range from zero to one. Hours are measured as working hours per month. Retired and working are binary variables indicating whether an individual's main source of income in a month comes from pension or work. Unless otherwise indicated, specifications include the baseline controls and fixed effects from Table 2. The baseline specification includes gender-by-birth-month-year fixed effects and month-year fixed effects, with standard errors clustered at the individual level. Standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

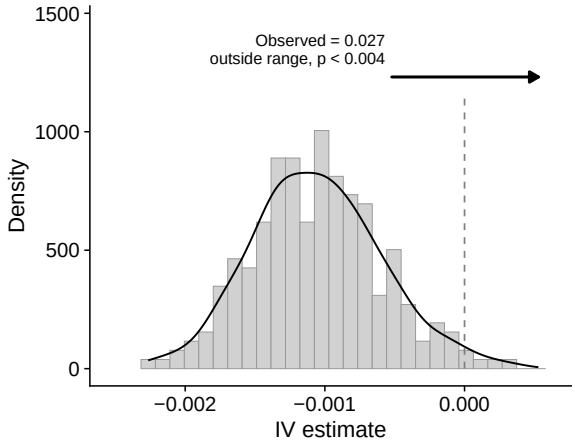
B.3 Detailed distribution of placebo estimates



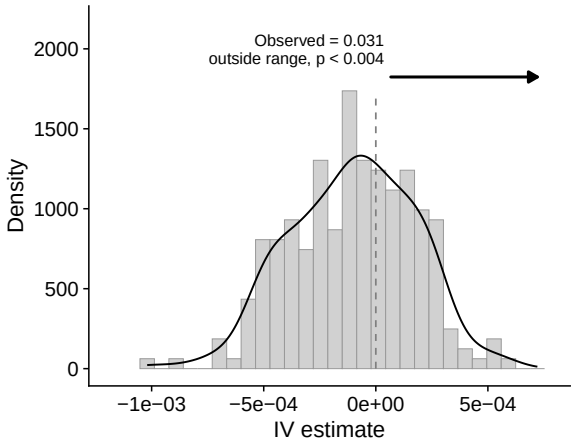
(a) Workplace network: hours



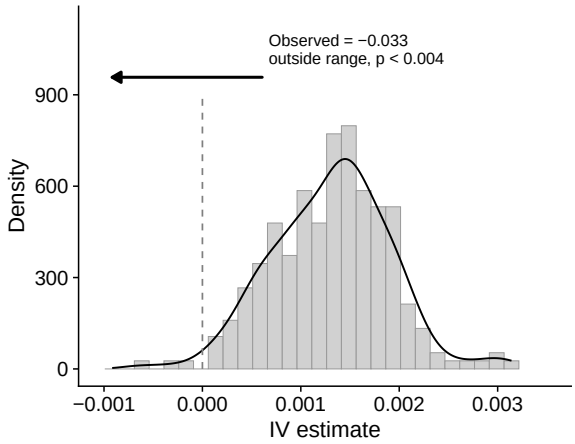
(b) Family network: hours



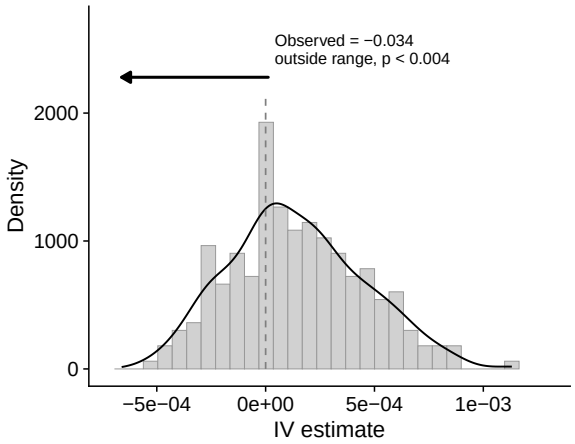
(c) Workplace network: retired



(d) Family network: retired



(e) Workplace network: working



(f) Family network: working

Figure 10: Placebo estimates from randomized peer networks

Notes: The histograms show the distribution of IV estimates obtained from randomized peer networks. The dashed vertical line marks zero. The solid vertical line marks the true estimate from the observed network when it falls within the displayed range. Peer-retirement shares range from zero to one.

B.4 Mapping retirement-status effects to retirement timing

This appendix derives the approximate interpretation of the retirement coefficients used in Section 7.2. Let T_i denote individual i's retirement month. Under the approximation that retirement is an absorbing state, the monthly retirement indicator can be written as

$$R_{it} = \mathbf{1}\{T_i \leq t\}, \tag{8}$$

such that

$$E[R_{it}] = P(T_i \leq t) = F_i(t). \tag{9}$$

Similarly, the share of retired peers can be interpreted as the empirical cumulative distribution of peer retirement timing, denoted by $F_{-i}(t)$.

Our baseline specification therefore approximately implies

$$\Delta F_i(t) = \beta \Delta F_{-i}(t). \tag{10}$$

For retirement timing measured in months, the change in expected retirement month can be written as

$$\Delta E[T_i] = -\sum_t \Delta F_i(t). \tag{11}$$

Substituting Equation (10) gives

$$\Delta E[T_i] = -\beta \sum_t \Delta F_{-i}(t) \tag{12}$$

$$= \beta \Delta E[T_{-i}]. \tag{13}$$

Hence, under the absorbing-state approximation, β can be interpreted as the approximate pass-through from a shift in average peer retirement timing to own retirement timing. A one-year shift in peer retirement timing therefore corresponds to an own-retirement shift of

12β months. This mapping is approximate because retirement is not fully absorbing in our data and because the baseline coefficient is assumed to summarize the peer effect over the relevant retirement window.